

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing reveals significant stability and reliability concerns with a 30% overall pass rate against the 90% target, indicating widespread endpoint failures across both API automation (27.5% pass rate) and backend smoke validation (33% pass rate) suites. The extensive failure pattern across 113 tests suggests systemic issues with API response handling, endpoint availability, or data processing logic rather than isolated defects, requiring immediate technical investigation to identify root causes in the backend services and establish a remediation path to achieve production readiness.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#65	14 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on failed endpoints to identify common failure patterns and backend service dependencies	Immediate	P1
2	Prioritize fixing high-impact smoke test failures to stabilize core revenue management functionality	3 days	P1
3	Review backend service logs and monitoring data to identify performance bottlenecks, timeout issues, or resource constraints	5 days	P2
4	Implement backend service health checks and circuit breakers to improve API reliability and error handling	1 week	P2
5	Establish daily backend stability reviews focusing on the most critical failed endpoints until 90% pass rate achieved	1 week	P2
6	Deploy incremental backend fixes with continuous validation against existing test suites to track improvement progress	2 weeks	P3

KEY FINDINGS

1	Backend services achieving only 30% pass rate with 113 failed tests out of 168 total, indicating critical system-wide issues
2	API automation suite showing 72.5% failure rate suggests widespread endpoint instability or incorrect response formatting
3	Smoke validation failures at 67% indicate core backend functionality and basic service operations are compromised
4	Consistent failure patterns across both test suites point to fundamental backend infrastructure or service configuration problems
5	60 percentage point gap from target pass rate requires comprehensive backend service review and stabilization

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation



29%

